

1) A primitiva de uma função é a antiderivada da função, ou seja, o processo inverso de uma derivada.

2) Integral Indefinida é a primitiva de uma função somada a uma constante C , onde C é um número pertencente aos reais.

3)

$$f'(x) = 3x^2 + (0 \cdot \cos(8x^3)) + \frac{1}{2} \cdot (-\sin(8x^3)) \cdot 24x^2 + \frac{12x}{6x^2}$$
$$f'(x) = 3x^2 - 24 \sin(8x^3) \cdot x^2 + \frac{12x}{6x^2}$$

4-a)

$$\int x^2(4x-8) dx = \int 4x^3 - 8x^2 dx$$
$$4 \int x^3 dx - 8 \int x^2 dx$$
$$x^4 - 8 \frac{x^3}{3} + C$$

4-b)

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$$
$$u = \sqrt{x} = x^{\frac{1}{2}}$$
$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$$
$$2 \int \sin u du$$
$$-2 \cos(\sqrt{x}) + C$$
$$2 du = \frac{1}{\sqrt{x}} dx$$

$$4-c) \int \cos x \operatorname{tg}(\sin(x)) dx \quad u = \sin(x)$$

$$\int \operatorname{tg}(u) du$$

$$\frac{du}{dx} = \cos(x)$$

$$du = \cos(x) dx$$

$$- \ln|\cos(u)| + C$$

$$- \ln|\cos(\sin(x))| + C$$

$$4-d) \int \frac{8x^3 + 1}{2x + 1} dx$$

$$\int \frac{\cancel{(2x+1)}(4x^2 - 2x + 1)}{\cancel{(2x+1)}} dx$$

$$4 \int x^2 dx - 2 \int x dx + \int 1 dx$$

$$\frac{4x^3}{3} - x^2 + x + C$$

$$4-e) \int \frac{\sin(\sqrt{x}) + 2\sqrt{x}}{2\sqrt{x}} dx$$

$$\int \frac{\sin(\sqrt{x})}{2\sqrt{x}} dx + \int \frac{2\sqrt{x}}{2\sqrt{x}} dx \quad u = \sqrt{x}$$

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$$

$$du = \frac{1}{2\sqrt{x}} dx$$

$$= \int \operatorname{se} u du + \int 1 dx$$

$$- \cos(\sqrt{x}) + x + C$$

$$4-f) \int \frac{2x}{4\sqrt{x^2-1}} dx = \int \frac{1}{4\sqrt{u}} du = \int u^{-\frac{1}{4}} du = 4 \frac{u^{\frac{3}{4}}}{\frac{3}{4}} + C = 4 \frac{(x^2-1)^{\frac{3}{4}}}{3} + C$$

$$u = x^2 - 1$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$\begin{aligned}
 4-g) \int x^3 \cdot e^{x^4} dx & \quad u = e^{x^4} \\
 \frac{1}{4} \int du & \quad \frac{du}{dx} = e^{x^4} \cdot 4x^3 \\
 \frac{1}{4} u + C & \quad \frac{du}{4} = e^{x^4} \cdot x^3 dx \\
 \frac{e^{x^4}}{4} + C &
 \end{aligned}$$

$$\begin{aligned}
 4-h) \int \frac{\sin(\ln(x))}{x} dx & \quad u = \ln(x) \\
 \int \sin(u) du & \quad \frac{du}{dx} = \frac{1}{x} \\
 & \quad du = \frac{1}{x} dx
 \end{aligned}$$

$$-\cos(\ln(x)) + C$$

$$\begin{aligned}
 4-i) \int \frac{\sin(2x)}{\cos(x)} dx & \\
 \int \frac{2 \sin(x) \cancel{\cos(x)}}{\cancel{\cos(x)}} dx & \\
 2 \int \sin(x) dx = -2 \cos(x) + C &
 \end{aligned}$$

4-j)

$$4-k) \int \frac{\cot g\left(\frac{1}{x}\right)}{x^2} dx \quad u = \frac{1}{x}$$

$$-du = \frac{1}{x^2} dx$$

$$- \int \cot g(u) du$$

$$- \ln \left| \operatorname{sen}\left(\frac{1}{x}\right) \right| + C$$

$$4-l) \int \operatorname{sen}^3(x) dx = \int \operatorname{sen}^2(x) \cdot \operatorname{sen}(x) dx$$

$$\int (1 - \cos^2(x)) \cdot \operatorname{sen}(x) dx$$

$$\int \operatorname{sen}(x) - \operatorname{sen}(x) \cdot \cos^2(x) dx \quad u = \cos(x)$$

$$- \cos(x) + \int -\operatorname{sen}(x) \cdot (\cos(x))^2 dx \quad du = -\operatorname{sen}(x) dx$$

$$- \cos(x) + \frac{\cos^3(x)}{3} + C$$

5-a)